

Prediction & Protocol Ledger

Long-Lived Harmonic State Induction in Photodiodes

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A. Claim (Predictive Statement)

Claim. A **sub-minute optical excitation** (< 60 s), consisting of **UV illumination combined with controlled angular modulation of concurrent visible light**, induces a **discrete harmonic electrical response** in a standard silicon photodiode that **persists for ≥ 30 days** after all illumination and motion cease.

The induced state is **absent prior to excitation**, does **not require continued optical or electrical input**, and exhibits **stable, non-random spectral structure** inconsistent with electronic noise or linear relaxation effects.

B. Protocol (Deterministic Recipe)

The following procedure is predicted to induce the long-lived harmonic state.

Device

- Photodiode: standard silicon photodiode (model to be specified)
- Condition: virgin, dark-stored, no prior UV exposure

Electrical

- Bias condition: as specified (or unbiased)
- Readout: high-impedance measurement (oscilloscope or equivalent)

Optical Excitation

- UV illumination:
 - Wavelength: 365–395 nm
 - Duration: 10–60 s
- Concurrent visible illumination:
 - Wavelength: specified (e.g. 532 nm)
 - Mode: continuous during UV exposure

Angular / Motion Modulation

- Geometry: relative angular or torsional modulation between light field, diode, and/or reflective boundaries
- Direction: specified (clockwise or counter-clockwise)
- Duration: simultaneous with UV exposure

Termination Order (Critical)

1. UV illumination OFF
2. Angular modulation OFF
3. Visible illumination OFF

No further excitation is applied after this sequence.

C. Expected Signature (Pre-Declared Observable)

If the claim is correct, the following must be observed:

1. **State Transition:** A transition from a flat or thermal baseline to a structured electrical signal following excitation.
2. **Discrete Spectrum:** Stable harmonic peaks with fixed frequency ratios (an n/k structure), reproducible across measurement sessions and persisting without continued illumination.
3. **Longevity:** Persistence of the signal for ≥ 30 days after excitation (with observed cases extending to multiple months).
4. **Stability:** Invariance of spectral ratios under power cycling of readout electronics, darkness, and environmental quiescence.

These features must appear without post-hoc tuning of analysis parameters.

D. Falsification Criteria (Explicit Kill Conditions)

This claim is false if any of the following occur:

1. **Virgin Failure:** A virgin photodiode subjected to the protocol does not enter a discrete harmonic state.
2. **Monotonic Decay:** The post-excitation signal exhibits only broadband noise or smooth exponential decay without discrete spectral structure.
3. **Non-Reproducibility:** The effect cannot be reproduced on two or more independent photodiodes using the same protocol.
4. **Continuous-Input Requirement:** The signal disappears immediately when illumination ceases, indicating no persistent state.

E. Scope Limitation

This ledger asserts **existence, inducibility, persistence, and falsifiability** of a long-lived harmonic state under the defined protocol. It does not assert a specific microscopic mechanism. Mechanistic interpretation is explicitly deferred.

F. Priority Statement

This document constitutes a **timestamped predictive commitment** that the described protocol will induce a long-lived, discrete harmonic electrical state in standard photodiodes. Any future experimental confirmation matching Sections B–D is covered by this priority claim.

End of Ledger